Elena Pagnin

Assistant Professor at **Chalmers** University of Technology

Personal Info

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Impact (Google Scholar)

No. publications: **30+**
No. citations: **500+**
h-index: **13**

Education & Titles

since 2022	**Assistant Professor** Tenure-Track (Forskarassistent)	Chalmers (SE)
2020 - 2022 Apr-Aug	**Associate Senior Lecturer** (Biträdande Universitetslektor)	Lund University (SE)
2019 - 2020 Feb - Mar	**Post-Doctoral Researcher** in Cryptography	Aarhus University (DK)
2014 - 2019 May - Jan	**Ph.D.** in Computer Science (Cryptography) Thesis Title: *"Be more and be merry: enhancing data and user authentication in collaborative settings"* *Supervisor: A. Sabelfeld*	Chalmers (SE)
2016 Aug 25th	**Licentiate** Degree of Engineering Thesis Title: *"Authentication under Constraints"* *Supervisor: A. Mitrokotsa*	Chalmers (SE)
2012 - 2013 Aug - Feb	**Project Officer** (paid Researcher Assistant position) Master thesis project under Prof. *F. Oggier* supervision	Nanyang TU (SG)

Recent Achievements (Grants & Awards)

2025	Nominated for the **Pedagogical Prize 2025** by the IT students; and my Cryptography course (TDA352/DIT352) appears in the recommended list of educational resources of **campus totalforsvar**	
2023-2024	Parental Leave (10 months)	
2023	**Co-PI** in a seed funding awarded by Chalmers' Areas of Advance on the topic *Towards a Multi-Layer Security Vision for Transportation Systems in the 6G Era*	**(0.5M SEK)**
2023-2027	**PI** of the prestigious Starting Grant awarded by the Swedish Research Council (VR starting) on *Progressive Verification of Cryptographic Schemes*	**(4M SEK)**
2020-2022	**PI** in a personal grant awarded to strategic research areas by the Excellence Center **ELLIIT**	**(1.6M SEK)**
2020	**Best Paper Award** for the full-version of *"Multi-Key Homomorphic Authenticators"* (extended abstract appeared in ASIACRYPT16) granted by the IET Information Security Journal. [This kind of Premium Award is given to recognize the best research papers published during the last two years]	

Supervision of PhD Students

(= I am the main advisor, = I co-supervise together with)

2024 - now	**Lucia Lavagnino**: *Transparency Logs* ( ) Planned PhD Defense: Nov 2029	Chalmers (SE)
2023 - now	**Adrian Perez Keilty**: *Progressive Verification of Cryptographic Schemes* ( ) Planned PhD Defense: 20 Aug 2028	Chalmers (SE)
2023 - now	**Hanna Ek**: *Advanced Properties for Digital Signatures* ( ) Planned PhD Defense: 31 Dec 2027	Chalmers (SE)
2019 - now	**Ivan Oleynikov**: *Privacy-Preserving Location Proximity Testing* ( ) Licentiate: 30 Sept 2022. Planned PhD Defense: December 2025	Chalmers (SE)
2022 - now	**Arthur Nijdam**: *Secure Machine Learning for the Medical Sector* (P. Stankovski-Wagner) Planned PhD Defense: September 2027	Lund University (SE)

Graduated Students

2025.03.07	**Joakim Brorsson**: *On the Trustworthiness of Trusted Third Parties* (P. Stankovski-Wagner)	Lund University (SE)
2023.03.21	**Martin Gunnarsson**: *Securing IoT Systems* (Prof. C. Gehrmann)	Lund University (SE)
2023.02.28	**Hadi Sehat**: *Dual Deduplication in multi-client setting and its applications* (Prof. D. Lucani)	Aarhus University (DK)

Selection of Professional Activities

09.06.2025	Member of the **grading committee** for the PhD Defence of *Joel Gärtner* on "Lattice-Based Post-Quantum Cryptography and Quantum Algorithms"	KTH (SE)
31.01.2024	Contributed to the **referral** to a new National law proposal on a Swedish electronic identification system *En säker och tillgänglig statlig e-legitimation* (SOU 2023:61)	
since 2021	**Program Committee Member** for: LatinCrypt25, ACNS25, LatinCrypt23, ACNS23, SEC@SAC22, ACISP21.	
since 2022	**Mentor** in the WISE-WWACQT mentorship program	
11.10.2023	**Seminar** speaker at ICT Area of Advance full-day seminar on Navigating the Cybersecurity Landscape on *Fortifying the Digital Fortress: Provably Secure Cryptography*	Chalmers (SE)

2023	**Invited Speaker** at AI NORDIC POWWOW on "Cybersecurity and AI for Company Security"	Lund (SE)
2023	**Seminar** speaker at the BARC center on *Progressive Verification for Cryptographic Schemes*	KU Copenhagen (DK)
2021-2022	**Scientific leader** at the EIT Department for the **SSF "SMARTY" grant RIT17-0035** (22M SEK)	Lund University (SE)
2022	**Invited Speaker** at the seminar series at *Protocol Labs*: Extended Threshold Ring Signatures	(remote)
2022	**Invited Speaker** at the CRC seminar series at *TII Research Centers*: Progressive and Efficient Verification	(remote)
2022	**Invited Speaker** at a National hearing of researchers on "Innovative Processes for Data Intergrity and Data Sharing" organized by the Swedish Ministery for Integrity Protection (Integritetsskyddsmyndigheten - IMY)	Stockholm (SE)
2022	**Invited Panelist** on "Allyship and Inclusion" at CRYPTO22	Santa Barbara (CA, USA)
2021	**Invited Speaker** at ELLIIT initiative on *Future-Oriented Research*: "Enhancing Data Authentication"	Lund (SE)
2021	**Participant** in the *Researchers' Grand Prix*: Security and Privacy in the Digital Era	Helsinborg (SE)
2020	**Invited Speaker** at *Framtidsveckan: AI, digitalisering och integritet – vad får vi för vår hälsodata?* Presentation and Panel Discussion on "Contact tracing apps and their security dilemmas"	Lund (SE)

Teaching Experience

2015-now	**Master's Theses**: **Supervisor** of 10, **Examiner** for 3.	Chalmers, Università di Milano, Lund University
2022-now	**Course Responsible** for the Masters level course *Cryptography* (TDA352/DIT352, 7.5hp)	Chalmers (SE)
2022	**Lecturer and Co-instructor** for the Masters level course *Advanced Cryptography* (EITN85, 7.5hp)	Lund University (SE)
2020 - 2022	**Lecturer and Course Responsible** for the Masters course *Advanced Web Security* (EITN41, 7.5hp)	Lund University (SE)
2021	**Course Responsible** for the PhD level course *Frontiers in Security Research* (EIT190F, 7.5hp)	Lund University (SE)

Selected List of Publications

CRYPTO25 Keilty, Aranha, **Elena Pagnin**, and Rodríguez-Henríquez: *"That's AmorE: Amortized Efficiency for Pairing Delegation"*. In: *Advances in Cryptology – CRYPTO* (2025).

PKC25 Baum, David, **Elena Pagnin**, and Takahashi: *"Universally Composable Interactive and Ordered Multi-Signatures"*. In: *International Conference on Practice and Theory of Public-Key Cryptography* (2025).

AC24 David, Engelmann, Frederiksen, Kohlweiss, **Elena Pagnin**, and Volkhov: *"Updatable Privacy-Preserving Blueprints"*. In: *Annual International Conference on Theory and Application of Cryptology and Information Security - ASIACRYPT* (2024).

SCN24 Baum, David, **Elena Pagnin**, and Takahashi: *"CaSCaDE:(Time-Based) Cryptography from Space Communications DElay"*. In: *International Conference on Security and Cryptography for Networks* (2024).

EuroS&P24 Nelson, **Elena Pagnin**, and Askarov: *"Metadata Privacy Beyond Tunneling for Instant Messaging"*. In: *IEEE European Symposium on Security and Privacy - EuroS&P* (2024).

CT-RSA23 Brorsson, David, Gentile, **Elena Pagnin**, and Stankovski-Wagner: *"PAPR: Publicly Auditable Privacy Revocation for Anonymous Credentials"*. In: *The Cryptographers' Track at RSA Conference* (2023).

ACNS22 Boschini, Fiore, and **Elena Pagnin**: *"Progressive and Efficient Verification for Digital Signatures"*. In: *International Conference on Applied Cryptography and Network Security - ACNS* (2022).

PKC22 Aranha, Hall-Andersen, Nitulescu, **Elena Pagnin**, and Yakoubov: *"Count Me In! Extendability for Threshold Ring Signatures"*. In: *International Conference on Practice and Theory of Public-Key Cryptography - PKC* (2022).

ESORICS20 Oleynikov, **Elena Pagnin**, and Sabelfeld: *"Where Are You Bob? Privacy-Preserving Proximity Testing with a Napping Party"*. In: *European Symposium on Research in Computer Security – ESORICS* (2020).

AC16 Fiore, Mitrokotsa, Nizzardo, and **Elena Pagnin**: *"Multi-key Homomorphic Authenticators"*. In: *Annual International Conference on Theory and Application of Cryptology and Information Security - ASIACRYPT* (2016).